

Tutorial Exercises – Series 4

Recursion

Exercise 1

Write a recursive algorithm that calculates the factorial of a given positive integer n .

Exercise 2

Write a recursive function $\text{sum}(n)$ that returns the sum of all integers from 1 to n .

Exercise 3

Write a recursive function to compute a^b (a raised to the power of b).

Exercise 4

Write a recursive function that returns the GCD of two integers using Euclid's algorithm.

Exercise 5

Write a recursive function that converts a decimal number into its binary representation.

Exercise 6

Write a recursive function that counts how many digits a positive integer has.

Exercise 7

Write a recursive function that returns the sum of digits of an integer.

Exercise 8

Write a recursive function to multiply two integers a and b using only addition.

Exercise 9

Write a **more efficient** recursive function to compute a^b using the rule:

$$\text{power}(a, b) = 1 \quad \text{if } b = 0$$

$$(\text{power}(a, b/2))^2 \quad \text{if } b \text{ is even}$$

$$a \times (\text{power}(a, b-1)) \quad \text{if } b \text{ is odd}$$

Exercise 10

Write a recursive function that returns the n -th triangular number. The n -th triangular number is the sum of all integers from 1 to n .

$$T(n) = n + T(n - 1)$$

Exercise 11

Write a recursive function that counts how many zeros a number contains.